
$$\mathcal{H}$$

$$E_{in}E_{out}$$

$$f^*\mathcal{H}$$

f^*	
h	f^*
$\mathcal{H}$	h

$$xyf^*S_n$$

$$f^*$$

$$S_n$$

$$h$$

$$\mathcal{X}x\in\mathbb{R}^d$$

$$\mathcal{Y}y$$

$$f^*f^*: \mathcal{X} \rightarrow \mathcal{Y}$$

$$S_n\{(x^{(i)},y^{(i)})\}_{i=1}^nf^*$$

$$hh\approx f^*$$

$$\mathcal{H}$$

$$\mathcal{H}$$

$$\rightarrow$$

$$\mathcal{H}$$

$$\mathcal{H}$$

$$(x,y)$$

$$h(x)\approx yh$$

$$x$$

$$x$$

$$E_{in}S_n$$

$$E_{out}\mathcal{D}$$

$$E_{in}E_{in}E_{out}$$

$$E_{in}\approx 0E_{out}$$

$$E_{in}E_{out}\approx E_{in}$$

$$S_nE_{in}\approx 0$$

$$E_{out}$$

$$\mathcal{H}f^*$$

$$E_{out}$$

$$f^*h\ f^*hF=maf^*F=0.98ma+0.02hf^*hf^*$$

$$S_nf^*h$$

$$\mathcal{H}$$

$$E_{in}E_{out}$$

$$E_{in}E_{out}$$

θ	
	$\theta \cdot x + \theta_0 = 0$
γ	

$$\theta \theta$$

$$d(d-1)$$

$$h(x) = (\theta \cdot x + \theta_0)$$

$$\theta \cdot x + \theta_0 > 0 \theta$$

$$\theta \cdot x + \theta_0 < 0 \theta$$

$$\theta \cdot x + \theta_0 = 0$$

$$\theta \theta$$

$$\theta$$

$$\theta = 0$$

$$x^{(i)} \hat{y} = (\theta \cdot x^{(i)})$$

$$\hat{y} = y^{(i)} \rightarrow$$

$$\hat{y} \neq y^{(i)} \rightarrow \theta$$

$$\theta_{new} = \theta_{old} + y^{(i)}x^{(i)}$$

$$+1\theta + x$$

$$-1\theta - x$$

$$\theta x \theta x$$

$$+1\theta x>90^\circ$$

$$\theta \leftarrow \theta + x$$

$$\theta x \theta_{new}$$

$$\theta x$$

$$\gamma$$

$$R$$

$$\gamma$$

$$k$$

$$k \leq \left(\frac{R}{\gamma}\right)^2$$

$$\gamma$$

$$+1-1$$

$$\theta = 0$$

$$x_1-1$$

$$\theta \cdot x_1 = 0$$

$$\theta \leftarrow \theta - x_1 \theta$$

$$x_2+1$$

$$\theta \cdot x_2 < 0 \rightarrow$$

$$\theta \leftarrow \theta + x_2 \theta$$

$$\theta$$

$$\theta$$

$$(\theta \cdot x) \theta$$

$$\theta_{new} = \theta + yx$$

$$(R/\gamma)^2$$

λ

λ	

$$Loss(y,z)=(0,1-yz),(z=\theta\cdot x)$$

$$yz\geq 1\rightarrow$$

$$0<yz<1\rightarrow 0\sim 1$$

$$yz<0\rightarrow 1$$

$$\|\theta\|$$

$$x\theta\cdot x=0\frac{|\theta\cdot x|}{\|\theta\|}|\theta\cdot x|\geq 1\frac{1}{\|\theta\|}$$

$$\frac{1}{\|\theta\|}\iff \|\theta\|^2$$

$$\frac{1}{2}\|\theta\|^2$$

$$y^{(i)}(\theta\cdot x^{(i)}+\theta_0)\geq 1(\forall i)$$

$$J(\theta)=\underbrace{\frac{\lambda}{2}\|\theta\|^2}_{\lambda\lambda}+\underbrace{\frac{1}{n}\sum_{i=1}^n(0,1-y^{(i)}(\theta\cdot x^{(i)}))}_{\lambda\rightarrow}$$

$$\lambda\lambda$$

$$\lambda\rightarrow$$

$$\lambda\rightarrow$$

$$\|\theta\|\frac{1}{\|\theta\|}\theta\theta\cdot x=1$$

$$z=1$$

$$(0,1-yz)$$

$$\frac{\lambda}{2}\|\theta\|^2+$$

$$\lambda$$

$$J(\theta)$$

$$J(\theta)\theta^*$$

$$J(\theta)$$

∇J	
η	

$$S_n$$

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} J(\theta_t)$$

$$-$$

$$\eta$$

$$t \rightarrow t+1n$$

$$n = 1,000,000O(n)$$

$$i$$

$$\theta_{t+1} = \theta_t - \eta_t \nabla_{\theta} Loss(x^{(i)}, y^{(i)}; \theta_t)$$

$$O(n) \rightarrow O(1)$$

$$n$$

$$\mathbb{E}[\nabla Loss(x^{(i)})] = \nabla J(\theta)$$

$$\eta$$

$$\eta_t$$

$$\sum \eta_t = \infty$$

$$\sum \eta_t^2 < \infty$$

$$\eta_t = \frac{1}{t} \eta_t = \frac{C}{t+}$$

$$\rightarrow$$

$$\rightarrow$$

$$\rightarrow$$

$$O(n)$$

$$O(1)$$

$$\eta_t$$

$$\phi(x)$$

$$x\phi(x)$$

ϕ	
	x^2, x_1x_2

$$(0,0), (1,1)(0,1), (1,0)$$

$$(0,0)$$

$$z = x_1^2 + x_2^2$$

$$z \rightarrow$$

$$z \rightarrow$$

$$x\in\mathbb{R}^d\phi(x)\in\mathbb{R}^DD>d$$

$$\theta \cdot x + \theta_0 \rightarrow \theta' \cdot \phi(x) + \theta'_0$$

$$x_1^2,x_1x_2$$

$$x=[x_1,x_2]^T$$

$$\phi(x)=[1,x_1,x_2,x_1^2,x_2^2,x_1x_2]^T$$

$$\theta^T\phi(x)=0x_1,x_2$$

$$w_0+w_1x_1+w_2x_2+w_3x_1^2+w_4x_2^2+w_5x_1x_2=0$$

$$QD$$

$$D\approx d^Q$$

$$d=100Q=3$$

$$\rightarrow$$

$$\rightarrow p \gg n \rightarrow$$

$$\phi(x)\phi(x)\cdot\phi(y)$$

$$x_1x_2$$

$$x_1^2,x_2^2,x_1x_2$$

$$x_1\times x_2\times$$

$$\phi(x)\phi(x)$$

$$\phi(x)=x^2\phi(x)$$

$$\phi(x)$$

$$1,x,x^2,\ldots$$

$$\phi$$

$$\phi(x)$$

$$\phi(x)$$

	w,bx
	α

$$\sum_{\alpha}\alpha_i-\frac{1}{2}\sum_{i,j}y^{(i)}y^{(j)}\alpha_i\alpha_j({}^{(i)}\cdot{}^{(j)})$$

$$x({}^{(i)}\cdot{}^{(j)})$$

$$\phi(x)K(x,z)$$

$$x,z$$

$$x\rightarrow \phi(x)z\rightarrow \phi(z)$$

$$\phi(x)\cdot \phi(z)$$

$$K$$

$$K(x,z)=\phi(x)\cdot\phi(z)$$

$$\phi K$$

$$K(x,z)=(x\cdot z+1)^2$$

$$\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n \times \mathbb{R}^n$$

$$K(x,z)=\left(-\gamma\|x-z\|^2\right)$$

$$x,z$$

$$\gamma \gamma$$

$$e^x1+x+\tfrac{x^2}{2!}+\tfrac{x^3}{3!}+\ldots$$

$$K(x,y)=\frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}t^2}\cos(xt)\cos(yt)dt\right)$$

$$K$$

$$\alpha$$

$$u$$

$$h(u)=\left(\sum_{i\in SV}\alpha_iy^{(i)}K(x^{(i)},u)+b\right)$$

$$uK$$

$$K(x,y)=\frac{1}{2}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}t^2}\cos(xt)\cos(yt)dt\right)$$

$$(x,y)$$

$$x^2,y^2,xy\dots$$

$$x^{(i)}$$

$$\gamma$$



$$C\gamma$$

$$\gamma \rightarrow \rightarrow$$

$$\gamma \rightarrow \rightarrow$$

$$n > 100,000$$

$$K(x,z)=\phi(x)\cdot\phi(z)$$

$$x^2, xy$$

$$\overset{W_1}{\rightarrow} \quad \overset{W_2}{\rightarrow} \quad \overset{W_3}{\rightarrow}$$

$$Wx + b\sigma$$

$$a^{[l]} = \sigma(W^{[l]}a^{[l-1]} + b^{[l]})$$

$$\sigma y = x$$

$$h = W_1x$$

$$y = W_2h = W_2(W_1x) = (W_2W_1)x = W_{new}x$$

$$\sigma(z)=\frac{1}{1+e^{-z}}$$

$$0\sim 1$$

$$(z)$$

$$-1\sim 1$$

$$f(z)=(0,z)$$

$$\hspace*{10cm}$$

$$\left|\left|\begin{array}{l} -\infty\sim\infty\\ 0\sim 1\end{array}\right.\right|$$

$$\hspace*{10cm}$$

$$x$$

$$[2.5, 0.1, 4.8] \rightarrow [0.09, 0.01, 0.90]$$

$$\rightarrow \rightarrow$$

W

	→
	→

$Lw\frac{\partial L}{\partial w}Lw$

$$\frac{\partial L}{\partial w} = \underbrace{\frac{\partial L}{\partial y}} \cdot \underbrace{\frac{\partial y}{\partial h}} \cdot \underbrace{\frac{\partial h}{\partial w}}$$

δa^4a^0

$L = (y - wx)^2y = 10, x = 2, w = 3$

$\hat{y} = 3 \times 2 = 6$

$L = (10 - 6)^2 = 16$

wL

$$\frac{\partial L}{\partial w} = 2(y - \hat{y}) \cdot (-x) = 2(4) \cdot (-2) = -16$$

w

$$\left|\frac{1}{n}\sum (y-\hat{y})^2\right|-\sum y\left(\hat{y}\right)$$

$$2/n$$

$$\eta\eta$$

$$\eta_t=\eta_0/(1+kt)()$$

28×28784

	3×3

3×3

→

2×2

$\rightarrow [+ \ +] \times N$

$224 \times 224 \times 3 \rightarrow \cdots \rightarrow 7 \times 7 \times 512$

$\rightarrow \rightarrow$

64×64

3×3

h_t	

$$h_t h_{t-1} x_t$$

$$xy$$

$$h_t = (W_{hh}h_{t-1} + W_{xh}x_t + b)$$

$$h_t t$$

$$h_{t-1} t - 1$$

$$x_t$$

$$W_{hh}, W_{xh}$$

$$t$$

$$C_t$$

$$f_t \rightarrow$$

$$i_t$$

$$o_t$$

$$~~~~~$$

$$\rightarrow$$

$$= \sum \alpha_i h_i \alpha_i$$

$$~~~~~$$

$$\rightarrow$$

$$\rightarrow$$

$$~~~~~$$

y

y

μ	
r	
J	
	$x,xyxyx$

$$K\mu$$

$$J(r,\mu)=\sum_{n=1}^N\sum_{k=1}^Kr_{nk}||x_n-\mu_k||^2$$

$$Jr\mu$$

$$r\mu$$

$$_rJ\implies \mu_k$$

$$\mu r$$

$$_\mu J\implies \mu_k$$

$$J$$

$$(1,1), (2,2), (3,3)(100,100)$$

$$\mu(26,26)$$

$$(2,2)(3,3)(100,100)$$

$$\left| \begin{array}{c} O(N) \end{array} \right| \left| \begin{array}{c} O(N^2) \end{array} \right|$$

$$KJKJ$$

$$\leftrightarrow$$

	$\rightarrow\rightarrow$

$$\iff$$

$$J = \frac{1}{N} \sum_{n=1}^N ||x_n - \hat{x}_n||^2$$

$$x_n$$

$$\hat{x}_n$$

$$R$$

$$R \approx U \times V^T RN \times M$$

$$UN \times K$$

$$VM \times K$$

$$ui\hat{r}_{ui}$$

$$\hat{r}_{ui} = {}_u \cdot {}_i = \sum_{k=1}^K u_{uk} v_{ik}$$

$$\times \times$$

$$_{U,V}\sum_{(u,i)\in}(r_{ui}-{}_u\cdot{}_i)^2+\lambda(\|U\|^2+\|V\|^2)$$

$$UVVU$$

$$K=2$$

$$u_A=[5.0,0.2]$$

$$v_{Nb}=[0.1,4.8]$$

$$\hat{r} = (5.0 \times 0.1) + (0.2 \times 4.8) = 0.5 + 0.96 = 1.46$$

$$\rightarrow$$

$$k=1$$

$$UV$$

$$R \approx UV^T$$

$$S, A, P, R, \gamma$$

